AIS – AURELIA INDUSTRIAL SYSTEMS

AIS-100 USER MANUAL

Aurelia Industrial Systems (AIS)
AIS-100 User Manual

Compact Assembly System
Document ID: AIS-MAN-100
Version: 2.1
Effective Date: January 15, 2026
Category: User Manual
Language: English
Classification: Internal / Customer Reference

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1. Introduction

The AIS-100 is a compact automated assembly system designed for industrial manufacturing environments.

The machine is intended for automated assembly operations where consistent production, repeatability, and integrated safety controls are required.

The standard AIS-100 configuration provides a nominal maximum production capacity of **40 units per minute**.

Actual production performance depends on:

- Product configuration.
- Material characteristics.
- Tooling.
- Upstream equipment.
- Downstream equipment.
- Operator configuration.
- Environmental conditions.

The machine must only be operated by trained and authorized personnel.

2. Intended Use

The AIS-100 is intended for industrial assembly applications within the operating conditions specified in the applicable technical documentation.

Typical applications include:

- Small-component assembly.
- Automated part insertion.
- Component positioning.
- Repetitive assembly operations.
- Compact production cells.

The machine must not be used for purposes outside its documented operating capabilities without prior approval from AIS Engineering.

3. Safety Requirements

Safety is a mandatory requirement for operating the AIS-100.

Operators must:

- Receive appropriate machine-operation training.
- Follow the approved startup and shutdown procedures.
- Verify that safety devices are operational.
- Keep unauthorized personnel away from restricted areas.
- Report abnormal machine conditions.
- Follow site-specific safety procedures.

Operators must never:

- Bypass safety interlocks.
- Disable emergency-stop devices.
- Modify safety circuits.
- Enter restricted machine areas while hazardous motion is active.

- Continue operation when a safety-related alarm is present.

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4. Emergency Stop

The AIS-100 is equipped with emergency-stop devices.

Activating an emergency stop places the machine in a safety-controlled state.

After an emergency stop:

1. Identify the reason for activation.
2. Verify that the machine area is safe.
3. Check for visible mechanical or electrical abnormalities.
4. Reset the emergency-stop device only when it is safe to do so.
5. Follow the normal machine recovery procedure.

An emergency-stop reset does not automatically mean that the underlying problem has been resolved.

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5. Installation

Installation must be performed by AIS personnel or authorized technicians.

Before commissioning, verify:

- Machine positioning.
- Electrical connections.
- Protective earth.
- Safety devices.
- Emergency stops.
- Guard interlocks.
- Industrial network connections.
- Required auxiliary services.
- Environmental conditions.

The machine must be installed on a suitable industrial floor capable of supporting its weight and operating conditions.

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6. Electrical Requirements

The standard AIS-100 electrical configuration is:

400 VAC, three-phase, 50/60 Hz.

The customer installation must provide the appropriate electrical protection and grounding according to applicable local requirements.

Electrical installation and maintenance must be performed by qualified personnel.

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7. Startup Procedure

Before starting the machine:

1. Inspect the machine and surrounding area.
2. Confirm that no unauthorized personnel are inside restricted areas.
3. Verify that emergency-stop devices are released.
4. Check that safety guards are closed.
5. Switch on the main electrical supply.
6. Wait for the control system to initialize.
7. Check the HMI for alarms.
8. Verify machine status.
9. Perform the approved homing or initialization sequence.
10. Enable automatic operation only after all safety conditions are satisfied.

If an unexpected alarm appears during startup, stop the procedure and consult the applicable troubleshooting instructions.

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8. HMI

The Human-Machine Interface provides access to operational information.

Typical HMI information includes:

- Machine status.
- Production status.
- Active alarms.
- Error messages.
- Operating counters.
- Maintenance information.
- Diagnostic information.

Only authorized personnel may change configuration parameters.

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9. Normal Operation

During automatic production, operators should periodically monitor:

- Machine status.
- Production counters.
- Active alarms.
- Material availability.
- Safety status.
- Abnormal noise or vibration.
- Product quality.

The standard nominal maximum capacity is:

40 units/minute.

Operating continuously above the documented machine capacity is not permitted.

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10. Shutdown Procedure

For a normal shutdown:

1. Stop automatic production using the HMI.
2. Allow the current operation to complete where applicable.
3. Confirm that the machine has reached the defined safe state.

4. Disable automatic operation.
5. Perform the required end-of-shift inspection.
6. Switch off the machine according to the site shutdown procedure.

For maintenance or extended shutdown, follow the applicable energy-isolation procedure.

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11. Preventive Maintenance

The standard preventive-maintenance interval for the AIS-100 is:

Every 1,500 operating hours.

Maintenance activities may include:

- Safety-device inspection.
- Mechanical inspection.
- Lubrication.
- Sensor inspection.
- Electrical-cabinet inspection.
- Fastener inspection.
- Actuator inspection.
- Alarm-history review.

Maintenance must be recorded against the machine serial number.

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12. Operator-Level Troubleshooting

Before contacting Technical Support, the operator may perform approved basic checks.

Examples include:

- Checking whether an emergency stop is active.
- Checking that guards are correctly closed.
- Checking visible sensor conditions.
- Checking material availability.

- Checking the HMI alarm message.
- Restarting the machine only when the manual explicitly permits it.

Operators must not perform electrical or mechanical intervention unless appropriately qualified and authorized.

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13. Common Problems

Machine Does Not Start

Possible causes include:

- Emergency stop active.
- Safety guard open.
- Safety circuit not ready.
- Initialization incomplete.
- Active machine alarm.

Check the HMI and safety status before contacting Technical Support.

Production Rate Lower Than Expected

Possible causes include:

- Material feeding issues.
- Product configuration.
- Tooling configuration.
- Downstream equipment limitations.
- Machine alarms or warnings.

The nominal capacity of 40 units/minute should not be interpreted as a guaranteed production rate under every configuration.

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14. Error Handling

When an error occurs, record:

- Machine serial number.

- Error code.
- Machine status.
- Time of occurrence.
- Production impact.
- Recent operator actions.

If the same error repeatedly occurs after the approved troubleshooting procedure, contact AIS Technical Support.

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15. Technical Specifications

Parameter	**Standard Value**
Product	AIS-100
Machine type	Compact Assembly System
Nominal capacity	40 units/min
Power supply	400 VAC, three-phase, 50/60 Hz
Operating temperature	5–35 °C
PLC	AIS Control PLC-100
Communication	Industrial Ethernet, digital I/O
Preventive maintenance	Every 1,500 operating hours

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16. Maintenance Safety

Before maintenance:

1. Stop production.
2. Place the machine in a safe state.
3. Isolate energy sources according to the approved procedure.
4. Verify that hazardous energy has been removed.
5. Only authorized personnel may perform maintenance.

Never perform maintenance while hazardous machine movement is possible.

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17. Technical Support

When contacting AIS Technical Support, provide:

- Machine serial number.
- Product model.
- Error code.
- Machine operating hours.
- Description of the problem.
- Production impact.
- Actions already performed.

This information allows Technical Support to begin diagnosis more efficiently.

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18. Document Control

Version	Effective Date	Description
2.0	2025-01-10	Updated operating procedures
2.1	2026-01-15	Updated maintenance and troubleshooting guidance

Current approved version: 2.1